Review - Exponential Functions

An exponential function is any function of the form $f(x) = P_0 a^x = P_0 e^{rt}$, where $a > 0$.

- If $P_0 a^t = P_0 e^{rt}$, then $a =$ _____. (Recall: $a^{bc} = (a^b)^c = (a^c)^b$.)
- P_0 is the _____ of the function. In other words, P_0 is the _____ of the graph of the function.
- If $a > 1$ (or $r > 0$), then the function is _____.
- If $0 < a < 1$ (or $r < 0$), then the function is _____.
- A bank account that pays interest rate r, compounded continuously, will yield \$_____ if we leave \$$A$ in it for t years.

Review - Inverses

- A function is *invertible* if it passes the _____.
- If $y = f(x)$, and $f(x)$ is invertible, then _____ $= x$

Preliminaries: Inverting Exponentials

Question Is every exponential function invertible? How do you know?

On the last worksheet, you were asked to find the doubling time of a bank account with interest rate 5%, compounded continuously. This required you to solve the equation

$$2A = Ae^{0.05t}$$

for t (where A is the initial deposit). To do so, you plugged in various numbers for t until you got pretty close to the answer. This is a little unsatisfactory.

Recasting this in terms of inverse functions: If we define $P(t) = Ae^{0.05t}$, we wish to compute _____.

If a quantity grows (or decays) exponentially, finding the time at which it reaches a particular value involves understanding the *inverse* of an exponential function.

For reasons we will see a bit later, we really only need to define the inverse of $f(x) = e^x$. For convenience, we will also define the inverse of $f(x) = 10^x$.

Logarithms

If $f(x) = 10^x$, then we define the *logarithm base 10* of x to be $f^{-1}(x)$. In other words

$$\log_{10} x = c \Leftrightarrow 10^c = x.$$

Often, we leave off the 10 and just write $\log x$.

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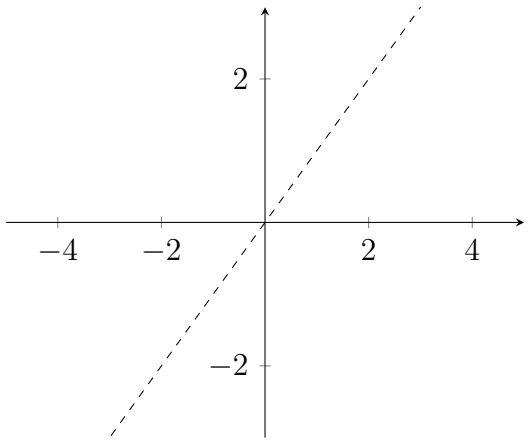
$$\log_e x = c \Leftrightarrow e^c = x.$$

For reasons we'll see next time, log base e is called the *natural logarithm* and is most often written $\log_e x = \ln x$.

- What are the range and domain of $\log x$ and $\ln x$?

- Why does it make no sense to find $\log 0$ or $\ln 0$? What about $\log x$ or $\ln x$ when x is negative?

- On the same axes, draw the graphs of 10^x and $\log x$, and of e^x and $\ln x$. (Hint: recall that the graph of the inverse of a function $f(x)$ is given by reflecting the graph of $f(x)$ in the line $y = x$, which is drawn for you).



- Do $\log(x)$ and $\ln x$ have vertical asymptotes? Where?

Properties of Logarithms

All the following properties can be deduced from the properties of exponents (eg $x^{a+b} = x^a x^b$):

$\log(AB) = \log A + \log B$	$\ln(AB) = \ln A + \ln B$
$\log\left(\frac{A}{B}\right) = \log A - \log B$	$\ln\left(\frac{A}{B}\right) = \ln A - \ln B$
$\log(A^p) = p\log(A)$	$\ln(A^p) = p\ln(A)$
$\log(10^x) = x$	$\ln(e^x) = x$
$10^{\log x} = x$	$e^{\ln x} = x$

Applications of Logs

Growth rates, Doubling times, and Half lives

1. □Regrade (a) Find the doubling time of a bank account that has 5% interest rate compounded continuously.

(b) If $P(x) = Ae^{rx}$, show that $P^{-1}(x) = \frac{1}{r}(\ln x - \ln A)$.

(c) Use part (b) to check your answer in part (a).

2. Suppose we start with 10 moles of radon-222, a radioactive element. After two days, □Regrade we find that there are 6.943 moles of radon-222 remaining.

(a) Without any computation: Is the half life more or less than two days? Explain.

(b) Let $N(t) = P_0 e^{rt}$ be the number of moles remaining after t days. Find P_0 and r.

(c) Find the half life of radon-222.

Continuous Growth and Decay Rates

If $f(x) = P_0 e^{rt}$, r is the *continuous growth rate*. It can be interpreted as the percentage rate at which function grows or decays, expressed as a decimal.

Examples The bank account in Question 1 above has a continuous growth rate of 5% per year, so $r = 0.05$. If you did Question 2 correctly, you should have found that $r \approx 0.1834$. So radon-222 decays at around 18.34% per day.

Inverses and Solving Equations

3. Find the inverses of the following functions.

□Regrade (a) $f(t) = 10(7)^t$

(b) $g(t) = 2\ln(t) + 5$

4. Solve the following equations. Give exact answers in terms of ln or log, not decimals.

☐Regrade

(a) $7^x = 2$

(b) $10^{2x+9} = 6 \cdot 7^{x-4}$

(c) $9^{7-2x} + 6 = e$ (Careful with this one! Before charging ahead, think about $e - 6$.)

Converting Between Exponential Forms

5. Let $f(t) = 7(2^t)$. Write $f(t)$ in the form $P_0 e^{rt}$.

☐Regrade

Converting Between Exponential Forms

Any exponential function can be written in two forms:

$$P(t) = P_0 a^t = P_0 e^{rt}.$$

This gives $a = e^r$, or $r = \ln a$. We will mostly use the latter form from now on, but you should be able to convert between the two.

More Problems

6. Use log rules to evaluate the following expressions without using a calculator. Show all your work. □Regrade

(a) $\log\left(\frac{1}{100}\right)$

(b) $\log\left(200^2\right) - \log(40)$

(c) $\log\left(\frac{1}{10,000}\right) + \log\sqrt{1,000}$

7. Let $x = \log A$ and $y = \log B$. Rewrite the following expressions in terms of x and y. □Regrade

(a) $\sqrt{\log(AB)}$

(b) $\frac{\log B}{\log A}$

(c) $\log\left(\sqrt{A}B^{-2}\right)$

(d) $\log\left(\frac{A}{B}\right)$

8. If possible, use the properties of logarithms to find exact solutions of the the following $\square_{\text{Regrade}}$ equations for x.

(a) $\log(1-x) - \log(1+x) = 2$

(b) $\log(10x-4) \cdot \log(16x^2) = 0$

(c) $\log(6x) + \log\left(2x - \frac{1}{3}\right) = 1$